

Arctic CDOM: a scoped halocline mixing diagnostic

P100-094 · Source-aware scientific reference. Supported as a scoped predictive extension; novelty and demonstrated industrial impact are not established.

Scenario and measured endpoint

Calibrated CDOM fluorescence channel in ppb. Contemporaneous calibrated CDOM fluorescence diagnostic using pressure,T1,S1; no other optical or oxygen response input. Good T/S flags 0, pump on,5–500 dbar. CDOM has no independent QC column.

Entire station casts 02,06,08 development (cast 04 has no eligible good-QC observations) leave-one-cast-out; casts 10 and 12 confirmation. Same expedition/instrument, spatial transfer not independent sensor calibration. Development contains 7,386 eligible observations in 3 groups. Confirmation contains 4,893 eligible observations in 2 groups. CDOM ppb is the publisher-calibrated fluorescence channel, not chemically measured DOC concentration. Quality control does not establish a universal conservative-mixing relationship. No target-based outlier removal.

Findings and executable relation

A continuous two-slope salinity relation with a thermal endmember correction improves held-cast CDOM prediction. It is compatible with changing water-mass mixtures across a halocline. CDOM fluorescence does not independently measure chemical DOC, and the model cannot separate mixing, biology or photochemistry.

The reference selected before confirmation is attempt-003:

$$\widehat{C} = a + b(S - 34) + c\max(S - 33, 0) + dT$$

C is the source-calibrated CDOM fluorescence channel(ppb), S is practical salinity(psu), and T is in-situ temperature(°C). The 33 psu hinge is fixed before fitting, not a newly identified phase boundary. a is in ppb, b,c in ppb/psu, and d in ppb/°C.

Coefficient	Exact fitted value
a	1.03476665527121
b	0.02073697165798316
c	-0.346510001524961
d	-0.01956237763234016

All numerical constants, transformations, alternative equations and fitted coefficients are executable in run.py and serialized in rules.json. Any algebraic consequence of this equation is an implication of the same fitted model, not a second independently validated finding.

Experimental method and evidence

5 substantive development attempts tested competing representations, controls and ablations. Fits use equal-group weighted least squares or declared robust loss, with all coefficients and radial-basis centers learned only inside each training fold. Selection minimizes mean group MAE, preferring fewer coefficients within 1% of the minimum. Separate confirmation checks the frozen states and performs no fitting. Baselines share permitted information.

Primary errors are in ppb; each group contributes equally regardless of row count.

Frozen model	Development MAE	Confirmation MAE	Worst confirmation group
reference	0.066903	0.0410217	0.0449196

Frozen model	Development MAE	Confirmation MAE	Worst confirmation group
constant	0.161173	0.176219	0.193638
flexible	0.0690253	0.0458915	0.0489194
salinity_mixing	0.0767451	0.0822193	0.0853564
attempt-001	0.0695723	0.0570697	0.0602191
attempt-002	0.0825597	0.063476	0.0658707
attempt-003	0.066903	0.0410217	0.0449196
attempt-004	0.080249	0.0556904	0.0608181
attempt-005	0.0705357	0.0629561	0.0648967

The frozen reference changes confirmation MAE by 10.6% relative to the strongest evaluated baseline `flexible` (0.0458915ppb); it wins on 2/2 whole groups. This is descriptive, not a population significance test. Rows within runs, days, weeks, profiles or casts are dependent. The complete group errors, bias, RMSE and 90th-percentile errors are in `evidence/by_group.csv`.

Falsification, limits and value

- A purely depth-based relation is weaker than the selected salinity-hinge relation on the reserved casts. - The threshold is a tested fixed representation, not evidence of a discovered universal salinity transition. - Report whole-group errors and ranges; no independent-row population CI. Temporal/spatial dependence and small number of sites/floats limit inference. - Public data and known physical parameterizations preclude automatic novelty certification. Predictive accuracy does not establish an industrial intervention or a new physical law.

Use complete casts and retain calibration/quality metadata. One planned development cast had every T1/S1 value flagged bad and was excluded by the prefit rule, reducing independent evidence to three development and two confirmation casts.

Reproduction and future evaluation

Run `python run.py` in this folder to verify allowlisted hashes and reproduce every saved confirmation prediction. This executes trusted bundled reference code; scoring an external prediction file should use the shared Principia evaluator without executing submitted code. Numerical alternatives need not resemble this equation. Use the exact registered task, timing/input budget, units, eligibility and group weighting. New endpoints or information access require a separately reviewed task version.

All confirmation outcomes are now exposed to future users. Source-stage exposure is recorded in `task_spec.json`; the first-use internal freeze history is preserved in research history. Independent source data are needed for a new confirmation campaign.

Primary sources and existing analyses

- <https://zenodo.org/records/17530624> — Dataset creator documents calibrated downcast sensors, paired packs and flags. CDOM is reported in ppb; no chemically measured DOC response or CDOM quality flag is supplied. - <https://pmc.ncbi.nlm.nih.gov/articles/PMC5034254/> — Primary Chukchi Sea study reports that optical DOM and DOC need not correlate simply with salinity; multiple water masses matter. Different region and date, so it informs falsification rather than validating this expedition.